Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

tudou

Selected Domain

Domain 1 - Future-Ready Workforce

Selected SDG Track (1–17)

8 - Decent Work and Economic Growth

Project / Solution Name

Pivott

Name (Leader)

Ho Xiang Ting

Email & Phone (Leader)

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University Affiliation(s)

Xiamen University Malaysia

Number of Team Members

2 - Ho Xiang Ting and Zheng Zimu

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.1. The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Clerical and finance workers face changing tasks as AI tools spread, but a job title alone does not show which parts of a working week could change. A Malaysian worker choosing an adjacent role or training course needs a task-level view of exposure, transferable skills and the effort needed to close a gap. Generic forecasts can overstate certainty. Public occupation and job data also have limits: O*NET describes US occupations, and Pivott's present Arbeitnow sample mainly covers Germany. Pivott addresses this information gap with visible sources and explicit uncertainty. It supports decisions about learning and decent work under SDG 8, with a skills link to SDG 4.4. It does not predict unemployment, guarantee a job or make hiring decisions.

2. EXECUTIVE SUMMARY & PROBLEM STATEMENT

2.2. The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

Pivott is a deployed career-planning prototype. A user selects or edits a role and tasks, rates skills, sees estimated task exposure under Slow, Normal and Fast scenarios, compares adjacent roles, and receives an ordered learning plan. A dated O*NET 31.0 occupation catalogue and Arbeitnow listing snapshot provide reference material; thin trend history is labelled insufficient rather than invented. The running product does not call a language model. Its scores use documented, deterministic TypeScript formulas, with an optional Python implementation for cross-checks. The public Next.js/React application uses route handlers and browser storage for guests; optional Supabase Auth and Postgres support account sync. Vercel hosts the main site. An official SDC sponsor-provided Replit deployment was used to host and debug another public instance at https://pivott.replit.app. The team applied workshop skills in GenAI-assisted development, database architecture and cloud deployment. A model-API coding agent drafted substantial backend code; the team wrote the contract and expected outputs, reviewed diffs, tested the implementation and resolved deployment issues. Cursor was the principal code-inspection and localhost workspace, and Claude Sonnet 5.5 assisted debugging. AI assisted the build, not the user's career scoring.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.1. Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Browser and local storage -> Next.js 16/React 19 interface and API routes -> deterministic TypeScript scoring engine -> committed O*NET 31.0 and dated Arbeitnow snapshots. Optional sign-in/profile sync -> Supabase Auth and Postgres with owner-bound row-level security. Optional separate FastAPI/SQLModel service supports imports, persistence and cross-engine checks; the public Vercel service does not require it. A reviewable GitHub Actions refresh obtains O*NET and Arbeitnow data and updates snapshots. Vercel hosts the main web app. Official SDC sponsor-provided Replit hosted and helped debug a second instance. No sponsor GenAI API is called by the running product. Development tools were a repository coding agent via a model API, Cursor for code inspection and localhost previews, Claude Sonnet 5.5 for debugging, and ChatGPT/Codex for research, documentation and the pitch deck. The team must confirm the exact model behind the repository coding agent before final sign-off.

3. TECHNICAL ARCHITECTURE & IMPLEMENTATION PLAN

3.2. Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We began with a synthetic finance-administration persona and the question a user needs answered: which tasks could change, which skills transfer, and what learning step follows? The backend developer wrote the API contract and data model before asking a coding agent for one bounded task at a time. The agent drafted routers, a Python scoring port, schemas and tests; the human developer set the canonical values, dataset boundary and authentication rules, reviewed each diff, ran tests, fixed a PostgreSQL verifier and CI drift, and resolved deployment routing. Slow, Normal and Fast scenarios show assumptions instead of claiming a forecast. The team checked local hosts in Cursor, used Claude Sonnet 5.5 to help diagnose bugs, and used sponsor-provided Replit for deployment and runtime debugging. The public prototype is accessible, but no real-user outcome or Malaysian demand trend is claimed. Phase 2 should test the workflow with Malaysian workers and improve local source coverage.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess Innovation and Creativity as well as Feasibility of build. Vague or unsupported projections may weaken your case for advancing to Phase 2.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Primary: UN SDG target 8.5, productive employment and decent work. Supporting: target 4.4, relevant skills for employment. Pivott contributes information for planning; it has not measured employment or educational outcomes. Sources: https://sdgs.un.org/goals/goal8 and https://sdgs.un.org/goals/goal4.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	As of this submission, 0 Malaysian pilot users have been evaluated and 0 real users have a measured Pivott plan completion. This is the product's measurement baseline (n=0), not a claim about national employment or skills. The Aisyah walkthrough is synthetic and does not count as a user outcome.

4. PRELIMINARY ENVIRONMENTAL & SDG IMPACT FRAMEWORK

Metric Category	Your Response
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided*	Share of Malaysian pilot participants who, without facilitator help, complete a task map, identify at least one skill gap and select or export a learning plan within 15 minutes. Report completions/participants, time and failure reasons. A completed plan is a usage outcome, not proof of employment impact.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed Phase 2 acceptance target: at least 70% (7 of 10) recruited Malaysian pilot users complete that workflow within 15 minutes. Report the actual sample and result even if the target is missed. This percentage is a pilot usability target, not a forecast of jobs created or an official SDG indicator.

5. ELIGIBILITY & READINESS CHECKLIST

Tick all boxes that apply before submitting.

- [x] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [x] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [x] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [ ] All fields in this template have been completed within the word limits stated.

6. IP & ORIGINALITY DECLARATION

6.1. Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

None, per team confirmation.

6. IP & ORIGINALITY DECLARATION

6.2. Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Next.js 16 / React 19	Code - web framework and UI	MIT
FastAPI / SQLModel	Code - optional backend	MIT
O*NET 31.0	Public occupation dataset	CC BY 4.0; attribution
Arbeitnow	Public job-listing API/data	Provider terms; cited
UN SDG labels	Public goal names/identifiers	UN source cited

6. IP & ORIGINALITY DECLARATION

6.3. Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

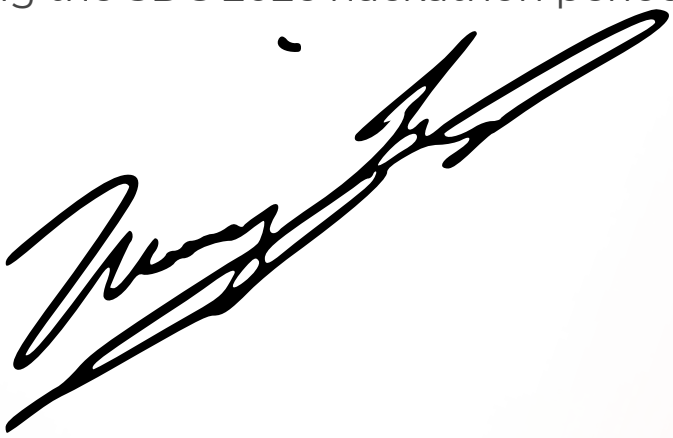
If yes, please describe the nature of the agreement and which member it concerns:

Neither team member reports an employment, internship or consulting IP agreement affecting this project.

7. DECLARATION

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]



[Date]

25/10/2026